

Babäk Firoozi Fooladi

Location: Espoo, Uusimaa, Finland
LinkedIn | GitHub | StackOverFlow
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Data Analyst | Real Estate Economist

Analytical and detail-orientated analyst and researcher graduate with a strong foundation in predictive modelling, econometrics, and data manipulation skills. Passionate about applying quantitative methods in practice, especially pricing analytics in commercial and built environments and construction. Effective communicator dedicated to delivering clear, data-driven insights that support international stakeholders and customer success.

Core Competencies & Technical Skills

- **Position related skills:** Reporting, academic and business communication, data analysis, data visualisation, presentation, collaboration
- **Data Analytics & BI:** Power BI, Dashboard Development, Data Visualization, Quantitative Research, Business Intelligence, Technical Documentation
- **Programming & Core Languages:** Python, R, SQL, Julia, STATA, Git
- **Data Engineering & Databases:** PostgreSQL, PostGIS, DuckDB, Pandas, Polars, GeoParquet, SQLite, MongoDB, REST APIs, Web Scraping, ETL/ELT Pipelines, Azure Functions, GCP, Azure
- **Spatial Data & GIS Analytics:** GeoPandas, Shapely, ArcGIS / ArcGIS Online, ESRI CityEngine, Spatial Econometrics, Cartography, Network Analysis
- **Data Science & ML:** Machine Learning, Econometric Modeling, Time-Series Forecasting (VAR, Co-integration, Univariate/Multivariate), OLS / IV-OLS, Panel Data, Survival Analysis, RAG, Agentic AI, LLMs/NLP

Professional Experience

DATA ANALYTICS & ENGINEERING

Self-Employed | Consultant

Jan 2025 – Present
Project-Based – Espoo, Finland

- Deliver end-to-end data engineering, spatial analytics, and business intelligence solutions tailored to client objectives and scalable growth.
- **Backend Architecture & Spatial Engineering (Apex-Heat):** Architected automated Python scripts using GeoPandas, Shapely, GeoParquet, and PostGIS to acquire, clean, geometry-correct, and model national spatial building datasets across Finland.
- **API & Data Pipeline Automation:** Designed and implemented API-driven data pipelines for automated spatial operations, evaluating data models for EU-wide scaling using domain expertise in real estate economics and zoning laws.
- **Pipeline Optimization & Tech Debt Reduction (Apex-Heat):** Spearheaded core codebase migration from Pandas to Polars, refactoring design patterns and introducing comprehensive test coverage to achieve an **86% performance improvement** on targeted calculations and a **60% overall application speedup**.

Ph.D. Candidate

Aalto University

Sep 2020 – June 2026
Full-time – Espoo, Finland

- Conduct quantitative research at the intersection of urban economics and real estate, applying econometrics and machine learning to housing market dynamics and spatial development.
- Architected and managed team data infrastructure, deploying a partitioned Data Lake and DuckDB-based querying workflows with full technical documentation for non-technical researchers.

- Engineered automated web scraping pipelines using Azure Functions and REST/ArcGIS endpoints to harvest, geocode, and store real-world property transaction data (MML, Ryhti, SYKE, KVKL).
- Modeled complex urban dynamics using time-series analysis, hedonic regressions, and time-to-event statistical modeling.
- Applied NLP and RAG architectures to process PDF/HTML land-use planning documents for structured feature extraction.

Research Assistant

Jun 2019 – Sep 2019

Aalto University Full-time – Espoo, Finland

- Collaborated with **MIT CityLab** on Agent-Based Modeling (ABM) to simulate urban mobility and parking pressure dynamics across campus infrastructure using the GAMA Platform.

Data Analytics Engineer

Sep 2020 – Dec 2024

Qissa kaupunkisuunnitteluanalytiikka Oy – Startup

Part-time – Espoo, Finland

- Designed and maintained end-to-end spatial data pipelines ingesting REST API data, performing geocoding and automated storage for urban planning analytics.
- Developed a spatial analytics dashboard with a custom network analysis backend, integrating HSL, HERE, and HSY APIs for real-time transportation routing and network evaluation.
- Processed multi-national census datasets (US, Finland, Sweden, Denmark) to calculate commuting pattern carbon emissions.

Planning Assistant

Nov 2018 – Jun 2021

WSP Finland Oy

Part-time – Helsinki, Finland

- Performed spatial data analysis and pedestrian network modeling to forecast route choices and post-development pedestrian traffic for municipal clients.
- Built interactive 3D city models using ESRI CityEngine and configured ArcGIS Online platforms for maritime spatial planning projects.
- Produced BI reports, spatial maps, and visualizations to communicate quantitative findings to multidisciplinary teams and executive stakeholders.

GIS Operator

Aug 2015 – May 2016

University of Tehran

Part-time – Tehran, Iran

- Collected, processed, and prepared **GIS datasets** to support urban designers in implementing and evaluating new urban design directives for Tehran.

Advisor & Assistant

Aug 2011 – Aug 2018

Airsa Dorsa Co. Ltd. – Family Business

Full-time – Karaj, Iran

- Managed financial planning, cost metrics, and operational workflows, leveraging accounting datasets to generate business intelligence reporting for executive decision-making.
- Double booking accounting, cost calculation, document management, and financial reporting for manufacturing operations.

Education

Aalto University

Doctor of Philosophy in Technology, Real Estate Economics

Espoo, Finland

Sep 2020 - Present

Aalto University

MSc. (Tech.) Urban Studies and Planning in Real Estate Economic

GPA: 4.1/5, Graduated with honours

Espoo, Finland

Sep 2018 - Jul 2020

languages

- Persian (Native)
- English (Fluent (C2))
- Finnish (Intermediate (A2-B1))

Publications:

- Effects of zoning on property values and impact assessment of changes in land use policy measures: [Source](#)

Certificates:

- Oxford Machine Learning Summer School, OxML 2023
- Urban Economic Association, UEA 2020 Lectures on Urban Economics

Community & Leadership

- **Speaker:** Spatial data analysis, spatial data engineering. DataTribe collective meetup, 2023
- **Contributor:** Helsinki Data Week volunteer (2024 & 2025)
- **Speaker:** "The Effect of Land Development Policies on Implementation of Housing Plans". 28th Annual European Real Estate Society conference, 22nd-25th June, Milan, Italy.
- **Speaker:** "The Impact of Public Land Development on Implementation of Housing Plans". AREUEA International Conference, 19th-22nd July, Cambridge, UK.
- **Speaker:** "Time limit: is it an effective way to encourage housing construction?". RYTS Science Slam, 25th of January, Helsinki, Finland
- **Speaker:** "Balancing financial constraints and public objectives: Assessing the impact of public land ownership on housing development duration". World Planning School Congress, 17th-21st July, Espoo, Finland
- DataTribe collective moderator
- **Guest Lecturer:** Science and Breakfast Staff Meeting, Introduction to Zotero, how to migrate from Mendeley. *October 2023.*
- **Guest Lecturer:** Guest lecturer in Spatial Planning and Traffic engineering (SPT) program, Aalto university. Data analysis and Visualization using PPGIS. *March 2023.*
- **Guest Lecturer:** Guest lecturer in Urban Studies and Planning (USP) program, Aalto university. How to use GIS in Urban studies and planning. *January 2023.*
- **Guest Lecturer:** Tutor in Urban Studio Challenge 1 (USP) course, Data analysis, GIS, planning and data visualization assistant. *September 2022.*
- **Guest Lecturer:** Guest lecturer in Sustainable Built Environment (SBE) course, Aalto university. GIS, Spatial Thinking and their use cases in sustainability and urban resilience. *November 2021.*
- **Guest Lecturer:** Guest lecturer in Sustainable Built Environment (SBE) course, Aalto university. GIS and its use cases in sustainability and urban resilience. *December 2020.*
- **Guest Lecturer:** Guest lecturer in Digital urban course at Aalto university. Teaching about data analysis and its uses in built environment, types of data and available information. *March 2020.*